And indeed they had. Penzias and Wilson went on to win the Nobel Prize in 1978 for the first observation of what became known as the *cosmic microwave background*.[fn10]

The cosmic microwave background, or CMB, went on to become one of the most important tools we have for studying the history of the universe. It's hard to overstate its significance, both as an astronomical data set and as a technological achievement. We can now collect, analyze, and map the glow of the hot early cosmos. The first thing it tells us is: the hypothesis that the early universe was one big inferno, glowing with heat, is completely confirmed.

But how do we know for sure that the background light we detect is actually from the primordial fireball, and not from, say, some collection of weird faraway stars or something? It turns out the light's spectrum—the way it's brighter or dimmer when measured at different frequencies—contains a dead giveaway.

Let's say you have a fireplace, and you stick a poker in the fire until it starts to glow red. That red glow isn't a property of the metal itself, it's a phenomenon that happens to anything that gets heated up (without bursting into flames). That glow is called "thermal radiation," and the color of it depends only on the temperature. Something glowing blue is hotter than something glowing red. In fact, if you could see infrared light, you'd see thermal radiation coming from people and warm food and sun-soaked sidewalks all the time. Human thermal radiation comes out at the low frequency of infrared light because we're much cooler than open flames, unless things are going very badly for us.

The color you see isn't the entirety of the light produced, though. Aside from lasers, anything that produces light makes it at a spread of different frequencies (or colors), and the color it appears to the eye is just the color where the light is most intense. (This is why incandescent light bulbs are hot to the touch: even though much of the light they produce is visible, a lot of extra light is produced in the infrared part of the spectrum, which makes them feel hot.) For any thermal radiation, including that emitted by pokers and people and the little blue flames on gas stovetops, the intensity of the light changes with frequency in exactly the same way. The light is brightest at some peak color depending on the temperature, and it dims quickly for colors on either side. Plotting out on a graph how the intensity rises and falls with frequency gives you a shape we call the *blackbody curve*—a